



École Polytechnique Fédérale de Lausanne

Flico

A Wilder Dataset for Dating Historical
Pictures Based on Flickr Commons

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1. Abstract

Flico (from Flickr Commons) is a dataset of 1,893,415 historical pictures under open-access licenses with high quality date annotations. The year of the creation date of each picture has been carefully extracted by combining information from multiple sources: from partially accurate existing metadata, from picture descriptions, titles, and tags, as well as from visual cues present in the pictures themselves. This automatic combination of information from multiple sources makes it possible to have both very high quality annotations and large scale data. This dataset covers a gap in historical image dating. Existing datasets focus on photographs and therefore contain very few pictures before 1930. Flico vastly covers the 19th century and early 20th century with illustrations, photographs, paintings, posters, and portraits. This opens possibilities to study historical images through a wider scope.

1.1. Keywords

date estimation, historical pictures, self-validation, data quality, Flickr Commons

Contents

1. Abstract	2
1.1. Keywords	2
2. Introduction	4
2.1. Context	4
2.2. Related Work	4
2.3. Contribution	5
2.4. Comparison	7
3. Data	8
3.1. Institutions	8
3.2. Flickr API	8
3.3. Metadata	9
4. Methods	11
4.1. Problem Definition and Hypotheses	11
4.2. Method Overview	11
4.3. Visual Date Estimation Model	12
4.3.1. Data Filtering	12
4.3.2. Sampling Strategy	13
4.3.3. Image Resolution	15
4.3.4. Embedding	15
4.3.5. Training	16
4.3.6. Regression models	16
4.3.6.1. Ridge Regression	16
4.3.6.2. Linear Regression	16
4.3.6.3. Support Vector Regression	16
4.3.7. Benchmark	16
4.4. Textual Date Extraction	17
4.4.1. Algorithm Overview	17
4.4.2. Scoring	18
4.4.3. Example	20
4.5. Aggregation Algorithm	21
4.5.1. Combined Year	22

5. Results	23
5.1. Method of Evaluation	23
5.1.1. 1850-2026 Metadata	23
5.1.2. 1850-1999 Metadata	23
5.1.3. 1000-2026 Human Label	23
5.2. Visual Regression (Using the 1850-2026 Metadata)	23
5.3. Comparison with Qwen (Using the 1850-1999 Metadata)	25
5.4. Complete Evaluation (Using the 1000-2026 Human Label)	25
5.4.1. Metrics Used	25
5.4.2. Mean Absolute error	26
5.4.3. Median Absolute Error	27
5.4.4. 90th Percentile Absolute Error	28
5.4.5. Mean Error (Bias)	28
5.4.6. Median Error (Bias)	29
5.4.7. Coverage	29
5.4.8. Data availability	30
6. Discussion	32
6.0.1. Limitations	32
6.0.2. Addressing Limitations	32
6.0.2.1. Limited Label Quality Guarantee	32
6.0.2.2. Methods Won't Generalize to Other Datasets	34
6.0.2.3. Visual Classifier: Garbage In/Garbage Out	34
7. Conclusion	35
Bibliography	36

2. Introduction

2.1. Context

A large historical picture dataset with accurate year labels is valuable for multiple reasons. Firstly, it lays the foundation for future quantitative analysis of historical phenomena like the evolution of fashion trends, the rise of photography in prints, architectural and urban development, or the evolution of medical practices, etc. Secondly, the dataset can serve as training and benchmarking material for computer vision tasks like date estimations of historical images.

Several aspects are used in this study to characterize such a dataset and to reflect design choices and real world constraints:

1. Source: The public archives hosting the pictures.
2. Access: The copyright rules under which the pictures fall.
3. Type: The type of pictures included: illustrations, posters, maps, paintings, book scans, calligraphy, postcards, portraits, landscapes, floorplans, photographs of events, animals, plants, buildings, people, etc. Different date estimation methods might specialize in a certain type of document and will need a filtered dataset.
4. Time frame: The time period covered by the dataset. Note that photographic datasets are limited to the time period after 1826, following the invention of photography.
5. Filtering: The method used to filter out poorly labeled data.
6. Correction: The method used (if any) to correct missing or incorrect values.
7. Size: How many pictures are in the archive.
8. Distribution: How the pictures are distributed across the time frame.

These characteristics are used in Section 2.4 to compare existing datasets (presented in Section 2.2) with the Flico dataset (presented in Section 2.3).

2.2. Related Work

The branch of the research focusing on date estimation of photographs heavily relies upon the “Date Estimation in the Wild” (DEW) dataset from Müller et al., 2017 [1]. It consists of 1,029,710 Flickr images, from 1930 to 1999. It contains a high diversity of concepts; city scenes, family photos, nature, historical events including elements considered as noise like close-ups of plants or animals as well as historical documents. For each year studied, the authors collected pictures by querying the Flickr API with three constraints. Firstly, the image is a photograph. Secondly, the picture’s `date_taken` metadata is between the 1st of January and the 31st of December of the given year. Thirdly, the year itself appears somewhere in the title, description, or tags of the picture.

DEW served as benchmark for date estimation models like Molina et al., 2021 [2]

DEW-B (B for balanced) was proposed by FNet et al., 2024 [3] as an extension of DEW. 414,877 pictures were added from Flickr [4], Europeana [5], and Laion5B [6] to balance the number of pictures per year. The years 1930 to 1969 were less represented than later years in the DEW dataset. The data added was filtered using various methods to correspond to photographs, “showing the daily life of the people.” The authors claimed that such photographs of people had higher reliability than more natural photos depicting landscapes, animals, plants, etc. The types of pictures were limited to daily-life scenes by using zero-shot image-text similarity predictions. Data with poor labeling was filtered out by using both a

date estimation model trained on DEW, and by removing pictures where the label differed from the prediction by more than 15 years.

Out of the 1,029,710 DEW pictures, 155,971 are under open access licenses¹ and are accessible online², while the rest are under copyright restrictions.

The DEW-B data is accessed by contacting the paper authors.

The De Boer Scene Detection dataset³ was also used by Barancová et al., 2023 [7] to estimate the date of historical photographs. The dataset is 20 times smaller than DEW with 52,160 digitized historical press photographs from 1945 to 2005. Barancová et al. focused on a subset of the dataset made up of 39,866 pictures for which dates were available, limited to the period between 1950 and 1999. This data set was not curated for date estimation but for detecting scenes like a protest, a wedding, a shopping street, or a baseball game.

Other works focus on specific types of photographs, like Ginosar et al, 2015 [8] focusing on high school yearbooks, Salem et al. 2016 [9] focusing on human appearance.

2.3. Contribution

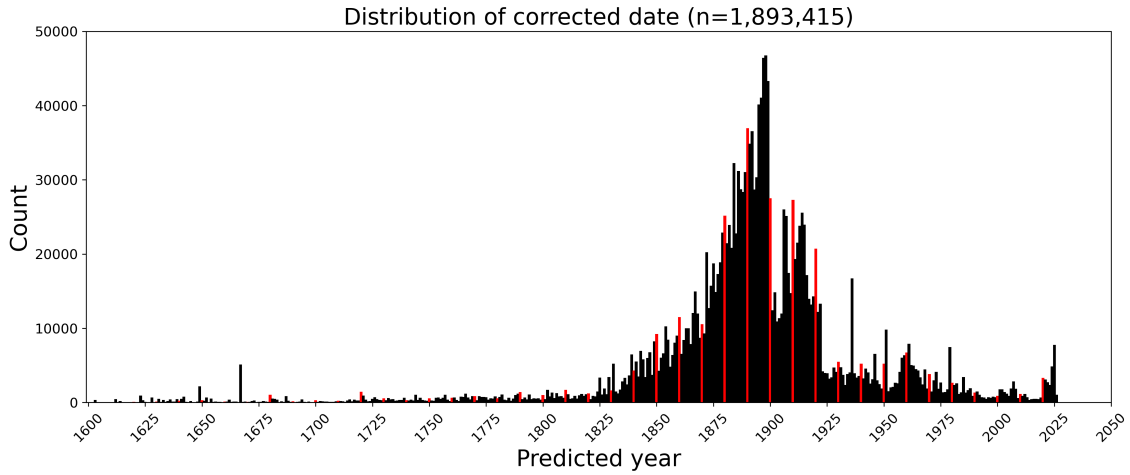


Figure 1: Date distribution per year in the Flico dataset

The author introduces Flico (for Flickr Commons): a dataset consisting of 1,893,415 pictures from Flickr commons. Every single picture is under open access license. The dataset contains pictures from the year 1000 to 2026 with most of the focus on the 19th and 20th century as visible in Figure 1. It contains various photographs as well as other types of visual documents: illustrations, etchings, paintings, posters, maps, etc. The author believes that the task of dating historical photographs is a bit self limiting because photographs are a relatively recent invention (1826). Because of how the technology was adopted over time, this really limits image dating possibilities to the 21st and 20th century. In fact, even the beginning of the 20th is not covered by DEW. Additionally visual dating methods that work on a certain type of photographs can work on equivalent documents of a different nature: Figure 2 shows an etching of a monument precise enough to be identified by photogrammetry and treated like a photograph of the same buildings (Zouzou⁴, Baroudi⁵, & Alameddine⁶, forthcoming).

¹<https://www.flickr.com/creativecommons/>

²<https://www.radar-service.eu/radar/en/dataset/toiMGdrQfZcWpnjy>

³<https://zenodo.org/records/7137452>

⁴https://github.com/Amine-Zouzou/final_version_flickr

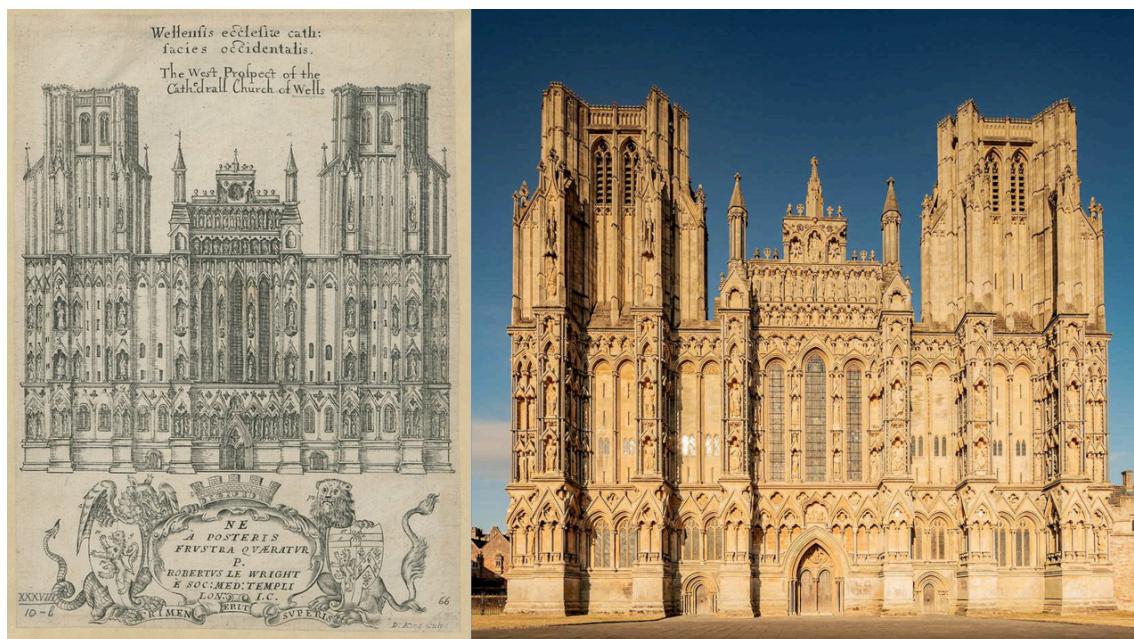


Figure 2: A 1656 etching⁷ of the Wells cathedral, and a modern picture of the facade.

For this reason, this dataset isn't restricted to photographs of scenes of daily life events.

Systematic errors in the Flickr `date_taken` metadata were identified and corrected. For example, some scanned photographs are mistakenly dated, with the year referring to the moment they were scanned and not the moment they were captured. This error mostly affects pictures taken after 2000, therefore the author doesn't think that it affects the quality of the labels in DEW. However, it means that a significant number of early 20th century pictures never made it into DEW because of their poor labeling inside Flickr. Other identified errors, like poorly labeled data was also identified and might have made it into DEW. The challenges of measuring the size of the problem, identifying missing/erroneous labels, and correcting them when possible is the main focus of this paper. The problems are addressed by extracting the date information from other fields like the title and the description. This effort on inferring missing/wrong labels automatically is the main contribution of this paper. Previous work (DEW and DEW-B) would simply filter out pictures with missing labels and assume that the existing labels were mostly correct.

In terms of methodology, this is a significant change that deserves to be discussed. History differs from other fields of computer vision because data cannot be generated/collected at will. Considering the overwhelming amount of knowledge about past human experiences that was never recorded, never understood, burnt in a fire, or never digitized, should invite to a deeper reflection when discarding data. Ignoring data that is poorly labeled greatly impacts the diversity of the final dataset because quality of annotation is correlated to the origin of the data. Each user (cultural institution in the case of Flickr commons) usually represents a certain local, regional, or national culture through the pictures they upload. If a user didn't fill up the `date_taken` field when uploading their pictures, their entire micro-culture is forgotten in the

⁵<https://github.com/ghassanbaroudi/flickr-filtering-db-integration>

⁶<https://github.com/kimoal276/flickr-project>

⁷<https://www.flickr.com/photos/britishlibrary/50264904536/in/photostream/>

eyes of the dataset, even though they might have systematically included the date in the description.

Flico is not meant to replace DEW. It is meant to cover areas that are overlooked by DEW: open access licensing, 19th century pictures, early 20th century pictures, diversity of picture types, diversity of cultures through correction of missing data.

2.4. Comparison

Table 1 is a summary of the differences between the state of the art datasets presented in Section 2.2 and the Flico dataset introduced in Section 2.3, using the characteristics from Section 2.1.

	DEW (2017)	DEW-B (2024)	Flico (2026)
Source	Flickr	DEW, Europeana, LAION-5B	Flickr Commons
Access	Largely copyrighted	Largely copyrighted	Open access
Type	Photographs	Photographs	Various types
Time frame	1930–1999	1930–1999	1000–2026
Filtering	Year as keyword	Prediction model	Prediction model
Correction	None	None	Textual extraction
Size	1,029,710	1,444,587	1,893,415

Table 1: Comparison of datasets for historical pictures dating

3. Data

The raw data used in Flico is 2,840,280 pictures and their metadata downloaded from Flickr commons via the Flickr API in March 2026. The data comes from 122 different institutions (The British Library, the U.S. Library of Congress, The Community Archive of Belleville and Hastings County in Ontario, etc) with various metadata about the pictures.

3.1. Institutions

The 122 institutions taking part in the Flickr commons program are Flickr users technically speaking. These institutions often use Flickr as a backup for public domain pictures in their collections. They often have their own metadata fields that may or may not coincide with the fields supported by Flickr. This creates a lot of diversity in the dataset in terms of what information is present for each picture, how this information is stored, and how consistent and accurate the information is.

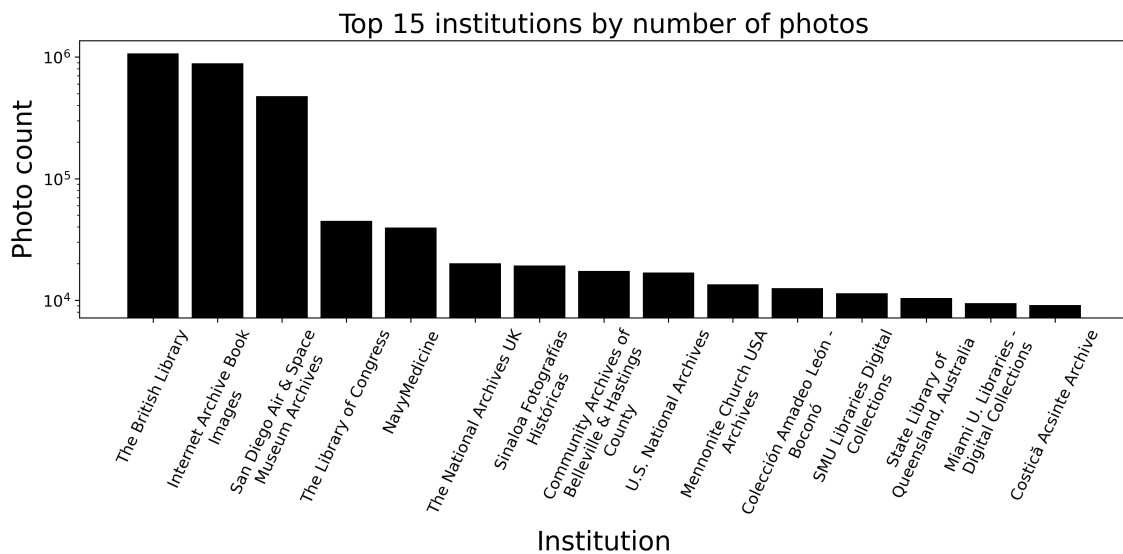


Figure 3: Number of pictures per institution in Flickr Commons

Figure 3 shows how the pictures in the dataset are spread across institutions. Three institutions dominate the dataset in terms of size. However, they don't necessarily have the most reliably annotated pictures, and a lot of the pictures from these institutions were not included in Flico.

3.2. Flickr API

Flickr offers an API service for a fee, allowing access to various metadata about users and their uploaded images. An API key was purchased and used to retrieve the relevant metadata about Flickr commons' users and their uploaded pictures. Note that Flickr hosts many more users and pictures that are sometimes licensed under public domain, but this study focuses on the commons specifically given their potentially higher historical value. The metadata retrieved for each picture combines automatic data that is generated by Flickr with data that is specified by the user when uploading their picture. The separation between the two types of data is a bit blurry but important for this study as it is often the distinction between intentional human labeling and automatic machine inference.

3.3. Metadata

Field	Type	Note	Example
id	Integer	identifies a picture	28207252096
owner_nsid	Text	identifies the institution owning the picture	109550159@N08
title	Text	Picture title	Farmers on main street, Saturday night. Iowa Falls, Iowa (LOC)
description	Text	Sometimes contains custom fields	Title: Histoire abregée des insectes Identifier: CUBiodiversity1154503 Year: 1762 Authors: Geoffroy, M. (Etienne-Louis), 1725?-1810 Subjects: Insects
date_upload	Timestamp	When the picture was uploaded to Flickr	2006-01-02 23:12:47
date_taken	Timestamp	When the picture was supposedly taken	1825-01-01-00:00:00
date_taken_granularity	Integer	A code indicating the precision of the date taken (0=to the seconds, 6=to the year)	8
date_taken_unknown	Boolean	Supposedly flagging pictures where the date_taken is unknown	True
tags	text	keywords about the picture	buildings mexico canals stereograph
url_n	text	the url of the image with its longest edge limited to 320 px	https://live.staticflickr.com/8477/8188498518_1fa7b3c1b5_n.jpg

Table 2: Picture’s metadata provided by the Flickr API that are used in Flico

The content of the metadata available with each picture is both user generated and automatically inferred from Flickr. Usually, Flickr will complete all the metadata it has information about and will let the user manually input the missing one if they wish to do so. For instance, when uploading a photo to Flickr, the `title` is inferred from the filename of the uploaded picture, and the user has the possibility to change it. This means that some pictures have very informative titles like “Vogue Model, Edmonton, Alberta, 1971” while others have very uninformative titles like “2604”. The `date_taken` field, that is most interesting for this study, can be either extracted from the EXIF metadata or manually inputted by the user. The author identifies this automatic EXIF inference is a systematic source of error for scanned pictures, that causes some `date_taken` fields to reflect the date of scanning rather than capture of the original photograph. Other fields are less tricky because they are either purely user inputs and have no default values: like `tags` and `description`, or they are purely handled

by Flickr internally: `image_id`, `owner_id`, `date_taken_granularity`, `upload_date`, etc. Fields that are pure user inputs like the `description` are empty when the user didn't fill them up.

Table 2 provides a list of all the relevant metadata fields that are used in Flico. These fields are provided by the Flickr API but little information is given about the source of the content of these fields. In particular, the exact way the `date_taken` is inferred, how the flagging with `date_taken_unknown` works, and the method to determine the precision of the `date_taken_granularity` are a bit mysterious. The methodology that was used to understand and exploit these fields in order to extract a consistent information is the focus of Section 4.

4. Methods

4.1. Problem Definition and Hypotheses

The general task of dating pictures was simplified to finding the year where the picture was taken. This aligns with annotations in the DEW dataset.

However, because the scope of this dataset is not limited to photographs, further questions are raised. A picture is taken in an instant, so there is no ambiguity about the year in which it was produced. A book however often takes more than a year to be produced. A lot of pictures from this dataset are photographs and illustrations taken from books. In such cases the date of publication of the book was prioritized as being the year from which the picture is from. There are a few specific cases for which this makes some of the data arguably irrelevant. For example, a book from the 18th century about the history of science and cartography can contain a 14th century map of the world that would be labeled as being an 18th century picture in this dataset.

The author argues that sacrificing a bit of precision in order to increase the amount of pictures labeled is beneficial when the information available is incomplete. For example, a picture taken during the first World War is better off as being dated from 1916 rather than being discarded. The rule decided upon by the author yields as follows: if partial information about a picture is provided and the possible years for the picture are narrowed down to a 5 year range, the midpoint of that range can be considered as a valid year for that picture. This compromise on precision is defended by multiple reasons: it allows for a greater number of pictures to be dated, it is within a reasonable range of 5 years, and more often than not a picture annotated in a 5 year range is annotated within that range because the range corresponds to a meaningful historical event (like a war), and that information is in a lot of cases precise enough to be relevant (being able to identify if a picture was taken during World War One is a useful task that won't suffer from this approximation).

4.2. Method Overview

To fully understand the methods used in this study, it is necessary to consider the timeline of the research that was conducted here. In particular, it is useful to mention that the author's knowledge about the quality of the dataset was very incomplete in the early stage. Each method used is presented here in chronological order. The results of said methods will also be discussed in this section if said result provided insights that motivated the design of the next method. It is important to keep in mind that the process of building the Flico dataset falls under the classification of interactive loop or human-guided automation as defined by Mladenovic et al., 2026 [10]: "An interactive loop implies back-and-forth exchanges between human and automation in the pipeline optimization process".

Initially, it was assumed that the `date_taken` field from the Flickr metadata was mostly correct. Manual inspection revealed that some of the data was in fact poorly labeled. The decision was made to leverage the two fields `date_taken_unknown` and `date_taken_granularity` to identify valid data. Filtering out pictures where `date_taken_unknown` is true greatly reduced the size of the dataset. It was therefore decided to attempt to predict the year for pictures where `date_taken_unknown` was true using a model trained on the images where the `date_taken_unknown` label was false. This is described in details in section Section 4.3. During this whole phase, the `date_taken` annotation was used as a ground truth label to evaluate the model's performance.

However, the evaluation of the visual model also revealed significant errors within the filtered data (`date_taken_unknown` is false). In particular, the data still suffered from the scanning date problem: an old picture is dated as being recent. The author's assumption was that this date might correspond in some cases to the date where the picture was scanned. In particular, it was assumed that this precision problem only concerned post-2000 pictures that were actually older and it was assumed that data before 2000 was accurate.

Realizing that a lot of the data was inaccurate encouraged the author to exploit other sources of information and use other metadata fields. The idea was to use what de Waal et al., 2011 [11] call an edit-rule to find inconsistencies within a dataset. The example provided by de Waal et al. is the one of checking that $Benefit = Revenue - Total\ Cost$. In our given dating task, we can extract the year where a picture has been taken from fields like the `title` or `description`, and compare it to the `date_taken`. This allows for a similar edit-rule, but with more nuances.

In order to extract the year from textual fields, a rule based Natural Language Processing was developed to extract date years candidates from textual fields. The rules were modified through iterations by comparing them to the `date_taken`. This is the focus of Section 4.4.

A final algorithm was developed to aggregate the information from the visual prediction, the textual date candidates, and the `date_taken` metadata. The algorithm labels the picture if there is sufficient coherent information. Instead of simply comparing the results and keeping pictures where the years extracted by the different method match, it was estimated that there were a significant number of cases where it was possible to identify which source was right, and use the date as final label. This is detailed in Section 4.5.

Around 600 pictures were manually annotated to form a benchmark on which the different methods and the aggregated result were compared.

4.3. Visual Date Estimation Model

In order to obtain a model capable of estimating the date of a picture from the image itself, a curated dataset with good image/date label pairs was prepared.

4.3.1. Data Filtering

From 2,840,280 pictures downloaded from Flickr commons, 1,363,199 contained a `date_taken` with the field `date_taken_unknown` set to false. It was assumed that this field was used to flag incorrect dates. Some data was identified as noise ($date_taken > 2026$) and was removed. Certain institutions were also filtered out for being unreliable (The British Library had 1,019,867 pictures dated from 2013 even though most of them were much older). Figure 4 provides an example of such a picture.

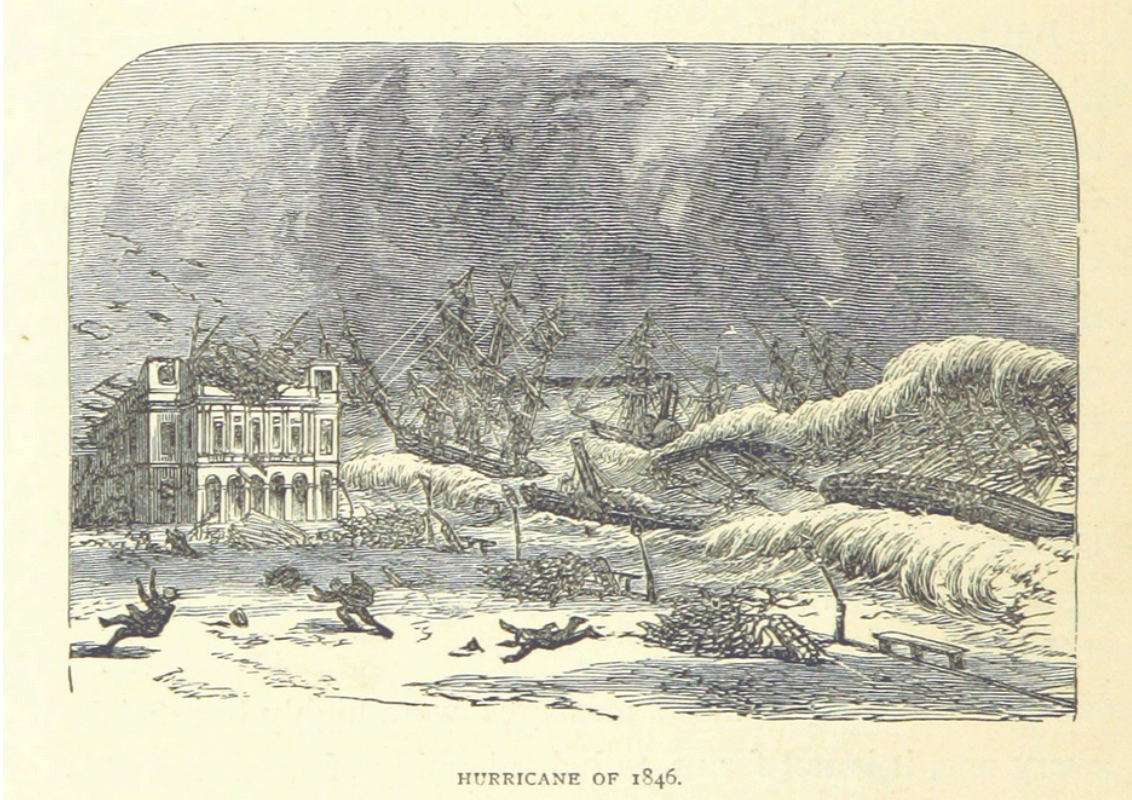


Figure 4: Picture from The British Library with wrong date_taken: 2013 (instead of 1873)⁸

Then in order to have a somewhat balanced dataset, 500 pictures were sampled for years between 1850 and 2026. There were very few labeled pictures before 1850 and some years were under represented, so the 500 pictures limit was chosen to keep a good class balance. The 500 pictures limit was chosen in order to balance data size (total amount of pictures in dataset) with coverage (effective size vs theoretical size). There are 176 years in the time span 1850-2026. If we sample 1000 pictures per year, the theoretical size of the dataset is 176,000 pictures. But, because some years have less than 1000 labeled pictures, the effective size of the dataset is 139,374. This corresponds to a coverage of 79%. A lower coverage means a less balanced dataset. Table 3 Shows how different sampling sizes affect coverage and size.

Sampled per year	Effective size	Coverage Ratio
3000	236,193	0.45
1000	139,374	0.79
700	108,959	0.88
500	83,400	0.95

Table 3: Impact of sampling size

4.3.2. Sampling Strategy

The sampling of the 500 pictures was not done randomly. Pictures with low granularity (more precise date_taken) were sampled in priority. This resulted in a relatively high granularity for most years before the 21st century as seen in Figure 5. The same strategy was performed to sample pictures in DEW [1]. A second criterion was used to sample pictures of the same year

⁸<https://www.flickr.com/photos/12403504@N02/11249514143>

and same granularity: institutional diversity was maximized. For example, if 50 institutions provide pictures with granularity 0 for a given year, 10 pictures are sampled from each institution. If an institution only has 7 pictures, the remaining 3 are sampled from other institutions with more than 10 pictures. This diversity makes the dataset more robust and limits cultural and geographical correlations with certain time periods.

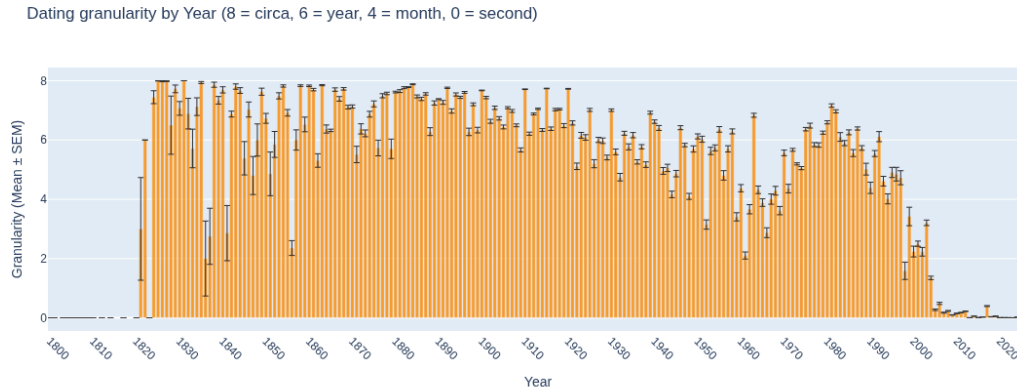


Figure 5: Average granularity per year in the sampled data

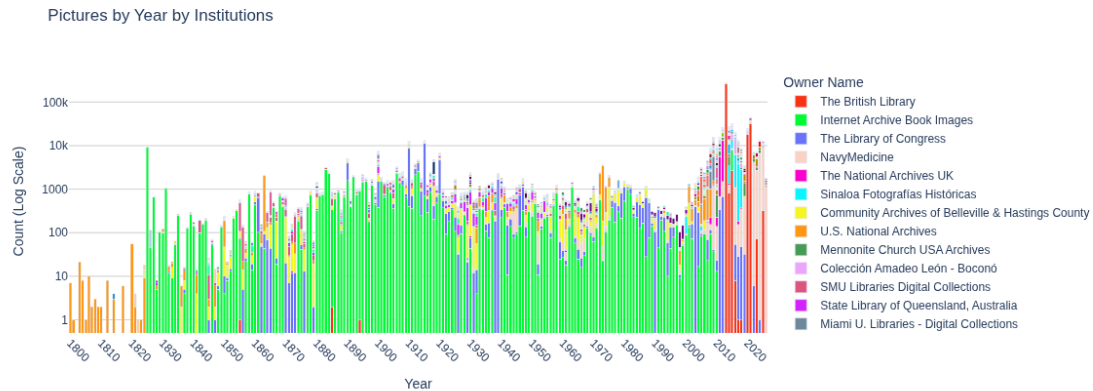


Figure 6: Distribution of pictures after filtering by date_taken_unknown=false, and before sampling

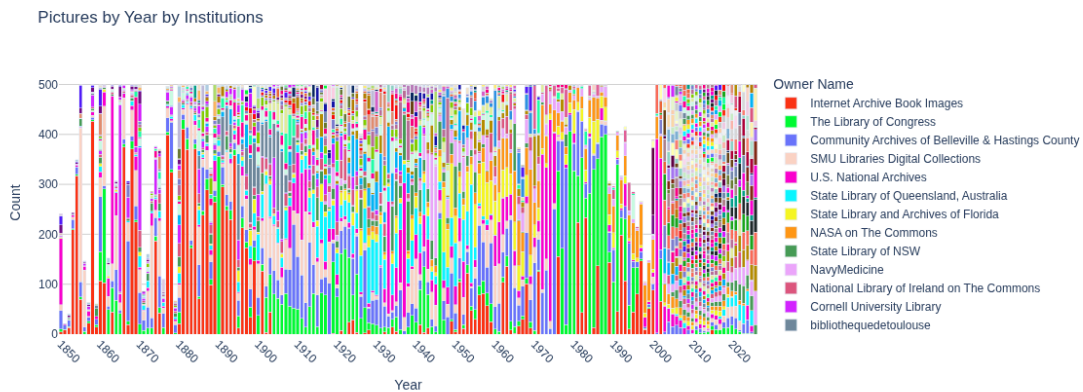


Figure 7: Distribution of pictures after sampling

Figure 6 and Figure 7 illustrate how the sampling strategy allowed for a more diverse and balanced dataset.

4.3.3. Image Resolution

Pictures were resized to all be of similar size (rescale such that longest edge is 320px). In practice, Flickr gives multiple download links for each image with variants of different sizes. Using the 320px variants instead of the original image is motivated by three practical reasons. Firstly, some high resolution scans are extremely large and slow to download. When processing pictures, having unreliable download times can slow down the entire process significantly and introduce a lot of challenges in terms of how to parallelize processing, handle timeouts, decide batch sizes... Secondly, working with smaller images also means that iterations are faster, and use less computing power. Thirdly, these images are not used directly, they are encoded in a more compact form using embeddings. Embeddings are less sensitive to image resolutions than other operations.

4.3.4. Embedding

Pictures were embedded into 768 dimensional vectors using Siglip 2, an embedding algorithm with visual and semantic aspects introduced in 2025 by Tschannen et al. [12].

Figure 8 shows a sample of 100,000 SigLip embeddings of pictures. Embedding vectors are normalized, and then reduced into a two dimensional space using the UMAP algorithm with 30 neighbors and a cosine similarity metric. The date used is the final corrected date from the aggregation algorithm. This plot is here just for illustration but possibilities to exploit such algorithm are discussed in Section 6.

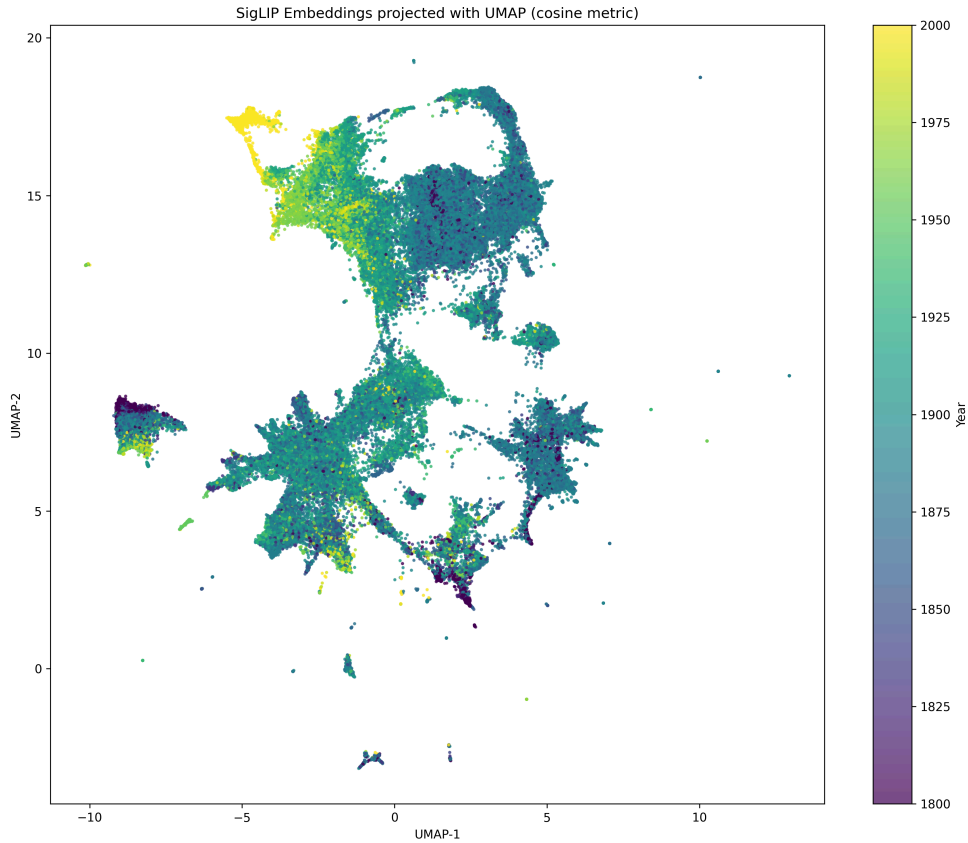


Figure 8: Projection of SigLip embeddings (random sample: 100,000 pictures)

4.3.5. Training

The 80k balanced dataset was split into a training and testing set using a 80%/20% split. Various regression methods were used to fit the data. They were evaluated against the `date_taken` label using Mean Absolute Error and Median Absolute Error. Some models were evaluated with multiple parameters. The best performing one was kept.

4.3.6. Regression models

Three regression algorithms were considered: linear regression, ridge regression and Support Vector Regression (SVR). These models were chosen because they can be trained quickly, with limited parameters, making iterations faster. The author initially intended to train more models (polynomial regression, random forest, gradient boosting), but efforts were redirected towards improving the quality of the training data instead through textual extraction of features from other fields (see Section 4.4).

4.3.6.1. Ridge Regression

Ridge regression was trained using 3 different values of the α parameter (the regularization strength): $\alpha = 10$, $\alpha = 2$, and $\alpha = 1$. Ridge regression tries to minimize the following loss function:

$$L = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^p \beta_j^2 \quad 1.$$

- $\sum_{i=1}^n (y_i - \hat{y}_i)^2$ is the residual sum of squares.
- β_j are the model coefficients.
- $\alpha \sum_{j=1}^p \beta_j^2$ is the L2 regularization penalty.

4.3.6.2. Linear Regression

Linear regression doesn't have any parameters. The sum of squares is minimized.

4.3.6.3. Support Vector Regression

Support Vector Regression (SVR) was trained using 3 different values of the C parameter: $C = 10$, $C = 50$, and $C = 100$.

Unlike linear and ridge regression, SVR does not attempt to minimize the prediction error of every training sample. Instead, it tries to fit a function such that prediction errors remain within a margin of tolerance (ϵ) while keeping the model as simple as possible. Observations lying outside this margin incur a penalty.

The C parameter controls the trade-off between model simplicity and training accuracy. Smaller values of C allow more prediction errors and result in a smoother model with stronger regularization. Larger values of C penalize errors more heavily, causing the model to fit the training data more closely at the risk of overfitting.

The implementation used the radial basis function (RBF) kernel, which allows the model to capture non-linear relationships between the input features and the target variable.

4.3.7. Benchmark

Because no model was previously trained on this dataset, a multi-modal large language model with chain-of-thought and vision abilities was used to predict dates from the test set. The model chosen is Qwen3-VL-235B-A22B-Thinking. It was prompted to predict the date of

pictures of the same resolution as the one used for SigLip embeddings. Listing 1 was used as a prompt.

Predict the year this image was taken as a four-digit integer (e.g., 2020).

Respond ONLY with valid JSON: {"year": <YEAR>}

Examples: - If 1995: {"year": 1995} - If 2023: {"year": 2023}

No explanations or extra text.

Listing 1: Prompt used for the benchmark

Both the Qwen benchmark and the visual predictor were evaluated through Mean Absolute Error and Median Absolute Error on the test set between the year 1850 and 1999 using the `date_taken` as ground truth label.

4.4. Textual Date Extraction

A rule based algorithm was developed to extract date information from the title, description and tags. The challenges of this algorithm are:

1. being able to extract valid dates, supporting a large amount of different formats, while filtering out codes and numbers that don't correspond to dates
2. being able to rank the various date candidates to differentiate the date of the picture's making from other dates that might be present in the description (the birth and death date of people in a portrait, date of some event related to the picture but not captured by it, etc).

Because this algorithm is meant to be compared with other dates instead of existing on its own, it doesn't output a single date but up to 3 candidates, ranked via a probability. It can therefore be used on its own by considering only its first guess, but it can also be used in combination with other methods. Details about how this is implemented in Flico can be found in section Section 4.5.

4.4.1. Algorithm Overview

Three fields were identified through manual exploration of the data as containing potential dates: the tags, title, and description. Here are the main steps of the algorithm:

1. The 3 fields are concatenated into one big string.
2. This string is preprocessed in order to normalize some notations (for instance: turning 'c1923' into 'circa 1923')
3. Some patterns are removed from the string in order to eliminate unwanted candidates (for instance removing 'The city was founded in 1854' that some institutions put in the description of all of their pictures from a given city, no matter when the picture was actually taken), or in order to date known publications or events (turning 'world war 1' into '1914-1918')
4. Book codes (like 'dertropenpflanze08ber') are substituted by their date of publication ('date year 1904'). For a lot of books, this was not necessary as the date was present in the description already. But, for some books or book series this date was lacking. This step was added after multiple iterations and was only done for large books or book series.
5. Regular expressions looking for 4 digit long sequences starting with 1 or 2 are used to extract candidates from the title, description, and tags (images from before the year 1000 are ignored).
6. The candidates are ranked based on the context in which they are found. For example, their score is increased if the keyword 'date' is on the same line while it is decreased if the keyword 'street' is on the same line (indicating that the 4 digit number is more likely a

house number rather than a year). The position of the candidate is also taken into account. For, example dates at the end of a line are more likely to correspond to the date of the document.

7. Scores are then normalized into probabilities.
8. Some institutions have been identified to rarely contain relevant date information while having descriptions that often contain 4 digits numbers. Dates extracted from these institutions are punished: their score is decreased, making it easier to filter them out later. For example the San Diego Air & Space Museum Archives often has 4 digits numbers corresponding to plane numbers that can be mistakenly identified as date candidates by the algorithm.

Further details for this algorithm can be found on GitHub⁹.

4.4.2. Scoring

Each candidate extracted from the textual fields is scored individually based on its context. The score corresponds to the sum of scores for the patterns from Table 4 that the candidate matches with, as shown in Equation 2.

$$s(c) = \sum_{p \in P_c} w(p) \quad 2.$$

where:

- P is the set of patterns from Table 4
- $w(p)$ is the score associated with pattern p
- $s(c)$ is the score of candidate c
- $P_c \subseteq P$ is the set of patterns matched by candidate c

⁹<https://github.com/nathanaelambert/flico/blob/main/src/trainer/date/description.py>

Pattern	Score
There is a bracket before the candidate	+13
There is a bracket after the candidate	+14
There is a parenthesis before the candidate	+7
There is a parenthesis after the candidate	+8
There is no other date on the line	+10
The candidate appears in the title	+5
The candidate appears in the tags	+12
There is an earlier date in the description	+6
There is a later date in the description	+6
The keyword 'date' appears on the same line as the candidate	+28
The keyword 'year' appears on the same line as the candidate	+28
The keyword 'produced' or 'created' appears in the same sentence as the candidate	+30
The keyword 'circa' appears in the same sentence as the candidate	+6
The candidate date finishes with a '0'	-7
'no date recorded' or 'ukjent datering' appear in the description	-50
The candidate date is older than 2030	-300
The candidate date is a serial number like '1524-AB-03'	-80
The keyword 'CO' or 'ref' appears in the same sentence as the candidate	-40
The keyword 'donated' or 'transfer' appears in the same sentence as the candidate	-50
The keyword 'number' or 'no.' appears in the same sentence as the candidate	-50
A '\$' appears in the same sentence as the candidate	-50
The keyword 'street' or 'road' appears in the same sentence as the candidate	-40

Table 4: Score per pattern

Then a score is attributed to each year for which there is at least one candidate by keeping the best score, and adding an affine score that depends on the number of candidates for that given year according to Equation 3.

$$S(y) = \max_{c \in C(y)} s(c) + 2|C(y)| + 100 \quad 3.$$

where:

- $C(y)$ is the set of candidates corresponding to year y
- $|C(y)|$ is the number of such candidates

The final score is then reduced for two specific institutions that often have date like numbers that don't actually correspond to dates. The two institutions are The San Diego Air & Space Museum Archive and Navy Medicine. The reduction of the score is done by multiplying the score by 0.88.

$$S'(y) = \begin{cases} 0.88S(y) & \text{for San Diego Air \& Space Museum Archive or Navy Medicine} \\ S(y) & \text{otherwise} \end{cases} \quad 4.$$

Then, the final (or reduced) score is normalised into a probability, as shown in Equation 5.

$$p(y) = \max\left(0, \min\left(1, \frac{S'(y)}{191}\right)\right) \quad 5.$$

The 3 years with the highest probability are kept and returned alongside their respective probabilities by the textual extraction algorithm.

4.4.3. Example



Figure 9: How did this cat make the news in 1924 ?¹⁰

Here are the title, description and tags that correspond to the picture shown in Figure 9:

- *Title:* How did this cat make the news in **1924**? (LOC)
- *Description:* Find out in our recent Picture This blog post: Coolidge's Cat: Out of the Bag!.
 ——— "Tige" the White House cat and pet of Mrs. Coolidge has been returned. **1924** March 25. 1 photographic print Summary: Benj. Fink, half-length portrait, standing, facing front, holding "Tige" the White House cat. Notes: • Caption continues: "Benj. Fink, guard at the Navy Dept. found Tige promenading around the Navy Bldg. and immediately returned him to the White House. "Tige's" disappearance was broadcasted by Wash. radio stations." • In album: 1 March to 30 April **1924**, v. 2, Herbert E. French, National Photo Company, p. 25, no. 29633. For more about the album, see: www.loc.gov/pictures/collection/coll/item/2004668683/ • Forms part of: National Photo Company Collection (Library of Congress). • Original glass negative may be available: LC-F8-29633 Subjects: • Coolidge, Grace Goodhue, —**1879-1957**—Animals & pets. • Pets—**1920-1930**. • Cats—Washington (D.C.)—

¹⁰https://www.flickr.com/photos/library_of_congress/43387018050/in/datetaken/

1920-1930. Format: • Photographic prints–**1920-1930.** • Portrait photographs–**1920-1930.**
Rights Info: No known restrictions on publication. Repository: Library of Congress, Prints and Photographs Division, Washington, D.C. 20540 USA, hdl.loc.gov/loc.pnp/pp.print
Higher resolution image is available (Persistent URL): hdl.loc.gov/loc.pnp/cph.3c31880

- Tags: Library of Congress, Tige, Cat, Presidentia Pets, Pet

The candidates extracted have been highlighted in bold. Table 5 shows the score assigned to each candidate. Table 6 shows the probability score assigned to each year. The algorithm only returns the first 3 years 1924, 1920, and 1930. Notice how the score for the 1957 candidate is higher than individual candidates for 1920 or 1930, but once the probabilities combine, because there are 4 candidates for each 1920 and 1930, the final probability score is higher for the date 1920 and 1930.

Candidate	Score	Source	Context
1924	56.99	title	cat make the news in 1924? (LOC)
1924	51.99	description	1924 March 25
1924	41.99	description	In album: 1 March to 30 April 1924
1879	34.62	description	1879 –1957–Animals & pets
1957	35.32	description	1879- 1957 –Animals & pets
1920	34.98	description	Pets– 1920 –1930
1930	34.98	description	Pets–1920- 1930
1920	34.98	description	Cats–Washington (D.C.)– 1920 –1930
1930	34.98	description	Cats–Washington (D.C.)–1920- 1930
1920	34.98	description	Photographic prints– 1920 –1930
1930	34.98	description	Photographic prints–1920- 1930
1920	34.98	description	Portrait photographs– 1920 –1930
1930	34.98	description	Portrait photographs–1920- 1930

Table 5: Score per candidate

Year	Probability
1924	0.853
1920	0.748
1930	0.748
1957	0.718
1879	0.715

Table 6: Probability score per year

4.5. Aggregation Algorithm

A simple algorithm was developed to combine the information from the date_taken field, from the description, tags, and title, and from the visual prediction. The algorithm’s goal is to identify correct values and exclude errors according to what was considered correct in Section 4.1. Because this algorithm is trying to establish the ground truth, it was evaluated on a subset of the data that was manually annotated. This is discussed in Section 5.

4.5.1. Combined Year

The aggregation algorithm relies on two values that were set by hand from manual observations.

- The first value is the minimum probability (See Equation 5) for a textual candidate (set to 0.69).
- The second value corresponds to the maximum range in years that is acceptable between two sources (set to 5 years). If two years are separated by less than 5 years, they are considered to be acceptably close.

Here is the list of cases that are considered to predict a valid date. They are in order, which means that if the first condition is satisfied, it is used to determine the date, if not the next condition is evaluated. If none of the conditions are satisfied, the data is not labeled.

1. If the `date_taken` and the most valid textually extracted candidate are the same year, this year is used to label the picture.
2. If the `date_taken` falls within the range of valid textual candidates and the range is small enough (less than 10 years), this year is used. (For example, the `date_taken` indicates that the year is 1917, and the two candidates 1914 with probability 0.75, and 1918 with probability 0.82 were extracted. The picture is labeled from 1917).
3. If the range between two valid candidates is small enough (less than 5 years), the picture is labeled from the average between the two candidates.
4. If the year predicted by the visual model falls within the range of valid textual candidates and the range is small enough (less than 10 years), this year is used.
5. If the year predicted by the visual model is the same as the most likely textual candidate (even if it scores under the valid threshold), this year is used to label the picture.
6. If the `date_taken` and the year predicted by the visual model are the same, this year is used.

Further details for this algorithm can be found on GitHub¹¹.

¹¹<https://github.com/nathanaelambert/flico/blob/main/src/trainer/date/combined.py>

5. Results

5.1. Method of Evaluation

Three different baselines were used to evaluate the different parts of the data pipeline along the development of the project. Each baseline reflects the understanding of the data at the time it was used.

5.1.1. 1850-2026 Metadata

The initial benchmark is the 0.2 split from the 80k set that is described in Section 4.3.1, Section 4.3.2, and Section 4.3.5. This was used to evaluate various regression methods.

5.1.2. 1850-1999 Metadata

The second benchmark consists of a restriction of the test set to pictures taken before 2000. This was done after identifying the “scanned-date problem” mentioned in Section 4.2. This reduced test set was used to compare the performance of the classifier with the performance of Qwen3-VL-235B-A22B-Thinking in a zero-shot regression task.

5.1.3. 1000-2026 Human Label

Considering that the `date_taken` field was identified to be at least partially incorrect, 600 pictures were annotated by hand to serve as a final benchmark to compare the quality of the metadata field with the textual extraction algorithm, the visual regression model, and the final aggregated year. 100 pictures were sampled for each of the six 50-years periods spanned by the dataset: pre-1800, 1800 to 1849, 1850 to 1899, 1900 to 1949, 1950 to 1999, post-2000.

5.2. Visual Regression (Using the 1850-2026 Metadata)

Model	Parameter	Train MAE	Test MAE
Linear Regression	—	24.98	25.45
Ridge Regression	$\alpha = 10$	24.92	25.40
	$\alpha = 2$	24.92	25.42
	$\alpha = 1$	24.92	25.41
SVR	$C = 10$	19.61	21.27
	$C = 50$	15.14	19.95
	$C = 100$	12.37	19.60

Table 7: Mean Average Error of regression methods

Table 7 shows the performance of the different regression models used. The performances are provided on the test and train set described in Section 4.3.5.

Figure 10 shows the confusion matrix for the SVR ($c = 50$) regression method. The big drop from the prediction target after the year 2000 actually highlights errors in the label rather than an issue with the regression. Pictures were manually sampled from that zone and identified as errors (example: Figure 11). Given that manual observations of the data revealed significant errors in the date taken label, an underfitting model seemed more appropriate and valuable. SVR with $C = 50$ was chosen as the final model, even though $C = 100$ has a better performance.

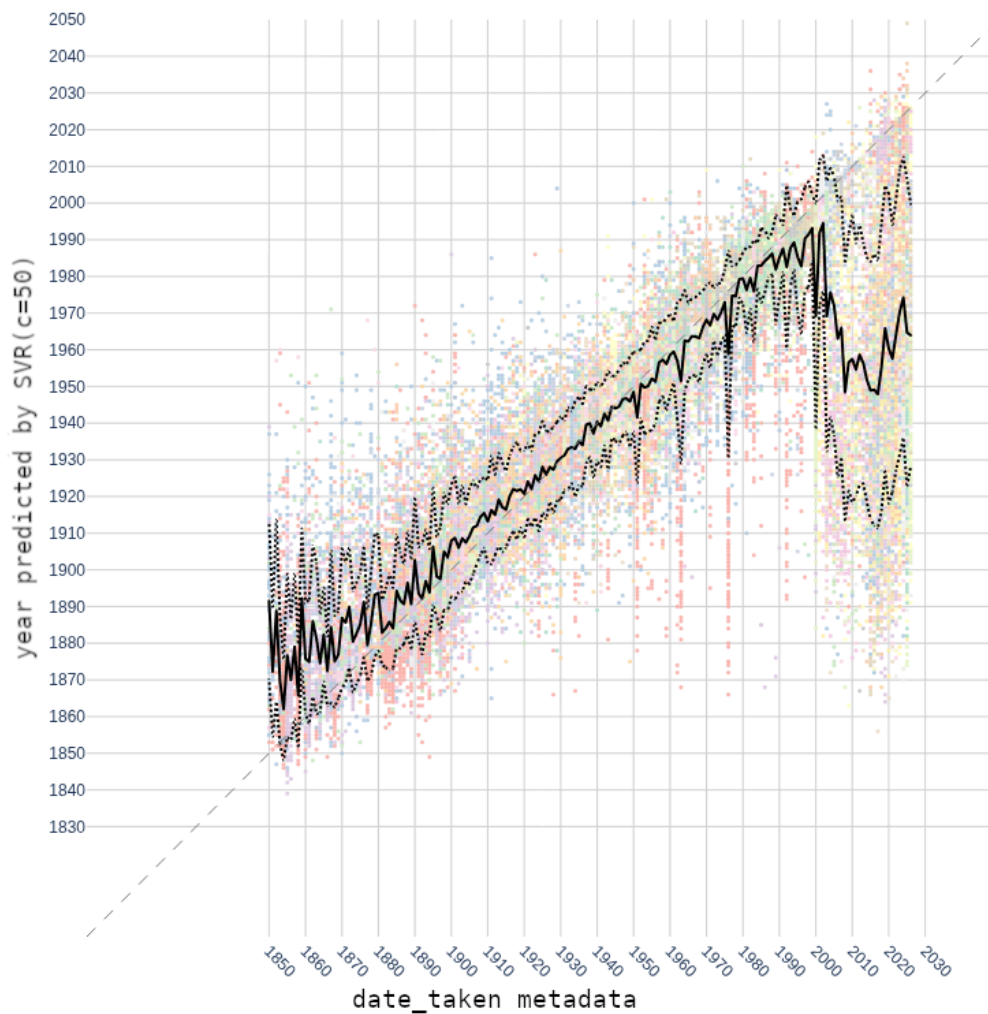


Figure 10: Confusion matrix of the year predicted by SVR($c=50$) vs the date_taken metadata



Figure 11: Picture with wrong date_taken: 2010 (instead of 1964)¹²

¹²<https://www.flickr.com/photos/36988361@N08/4991234643>

5.3. Comparison with Qwen (Using the 1850-1999 Metadata)

Given that the identified systematic error concerned pictures after the year 2000, the visual prediction model was evaluated on the period 1850 to 1999. The visual predictor was compared to the predictions made by Qwen3. The textual extraction model was also evaluated on that same data.

Model	Mean Absolute Error	Median Absolute Error
Qwen3	16.30	10.0
SVR(c=50)	11.45	8.0
Textual	7.71	0.0

Table 8: Comparison between Qwen, SRV50, and Textual extraction

This solidified the choice of using SVR(c=50) as the visual model in the pipeline, since it was good enough to beat a state of the art multi-modal large language model with visual and reasoning capacities.

5.4. Complete Evaluation (Using the 1000-2026 Human Label)

Since the aggregation algorithm uses the date taken metadata as part of its decision process in order to determine the actual year when a picture is from, there is no sense in evaluating the aggregated result against a baseline that relies on the date taken field as a source of truth. The aggregated result was therefore evaluated on a stratified sample of 600 manually labeled pictures. This is a small sample considering the size of the whole dataset.

5.4.1. Metrics Used

Several metrics were used to evaluate the different dating methods. Since predicting a year is a regression task, relying on a single metric can hide important properties of the predictions.

Mean Absolute Error (MAE) measures the average absolute distance between the predicted year and the ground truth year:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad 6.$$

where $\hat{y}_i$ is the predicted year and y_i is the manually annotated year.

MAE is easy to interpret because it is expressed directly in years. A MAE of 5 means that predictions are on average 5 years away from the correct date.

Median Absolute Error (MedAE) is the median of the absolute errors:

$$\text{MedAE} = \text{median}(|\hat{y}_i - y_i|) \quad 7.$$

Unlike MAE, the median is less sensitive to a small number of large mistakes. It provides information about the typical prediction error.

90th Percentile Absolute Error (P90AE) corresponds to the value below which 90% of the absolute errors fall:

$$\text{P90AE} = Q_{\{0.9\}}(|\hat{y}_i - y_i|) \quad 8.$$

where Q_p denotes the p -quantile of a distribution. In other words, $Q_{0.9}$ is the value below which 90% of the absolute errors are located.

This metric measures robustness. A low value indicates that even large mistakes remain relatively limited.

Mean Error (Bias) measures whether a method systematically predicts years that are too early or too late:

$$\text{Bias}_{\{\text{mean}\}} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \quad 9.$$

A positive value indicates a tendency to predict dates that are too recent, while a negative value indicates a tendency to predict dates that are too old.

Median Error (Bias) is defined as:

$$\text{Bias}_{\{\text{median}\}} = \text{median}(\hat{y}_i - y_i) \quad 10.$$

This metric captures the typical direction of the prediction error while remaining robust to outliers.

Coverage corresponds to the proportion of pictures for which a method produced a prediction:

$$\text{Coverage} = \frac{N_{\{\text{predicted}\}}}{N_{\{\text{total}\}}} \times 100 \quad 11.$$

A method with excellent accuracy but low coverage may be less useful in practice than a slightly less accurate method that can process the entire dataset.

5.4.2. Mean Absolute error

Table 9 shows the average prediction error of each method on the manually annotated benchmark.

The aggregated result consistently achieves the lowest MAE across all historical periods. For pictures produced before 2000, the error remains below 1.3 years in every period and below half a year for the periods 1800–1849 and 1950–1999. This demonstrates that combining multiple information sources is substantially more reliable than relying on any individual source.

The metadata `date_taken` field performs poorly before 2000. Errors range from 26 to 259 years depending on the period. This confirms that metadata alone cannot be considered a reliable source of truth for historical images. The largest errors occur before 1900 where scanned-date issues and cataloging inconsistencies are most common.

The textual extraction method achieves remarkably low errors. For the periods before 1850, the MAE is below two years and even reaches 0.01 years for the period 1800–1849. This confirms that institutions often store accurate date information inside titles and descriptions even when the Flickr metadata itself is incorrect. Performance decreases slightly during the twentieth century because descriptions often contain multiple historical dates that are harder to disambiguate automatically.

The visual model performs substantially better than the metadata field despite being trained on the metadata field. This indicates that the choice to keep an underfitting model was good and made it more robust. As expected, its errors are larger than those of textual extraction

because visual dating is inherently ambiguous, and because the model was not trained on pre 1850 pictures. Nevertheless, errors between 15 and 33 years across most periods remain informative enough to support the aggregation process.

The post-2000 period is the most challenging for the aggregated result. The metadata and textual extraction are often the most reliable sources of information for this period, but it is more difficult to compare them to obtain very accurate results. Scanned-date corrections become more challenging because the true date is already close to the digitization date, distinguishing between them becomes harder. Despite this difficulty, the aggregated result remains the best-performing method.

Period	Aggregated	Metadata	Text	Visual
<1800	0.21	259.23	1.77	230.96
1800–1849	0.08	87.45	0.01	82.95
1850–1899	1.07	70.38	14.88	32.98
1900–1949	1.25	46.23	23.35	15.41
1950–1999	0.44	26.60	25.07	16.35
≥2000	5.13	5.46	6.25	13.78

Table 9: Mean Absolute Error

5.4.3. Median Absolute Error

The median absolute error results in Table 10 provide an even clearer picture of the effectiveness of the aggregation strategy.

For every period, the aggregated result achieves a median absolute error of 0 years. This means that more than half of the manually annotated images are assigned the exact correct year. Such performance indicates that the aggregation rules successfully identify highly reliable situations where multiple sources agree.

The textual extraction algorithm also achieves a median absolute error of 0 years in every period. This highlights the fact that when explicit dates are present in the metadata fields, the extraction algorithm is usually able to identify them correctly.

In contrast, the metadata field exhibits large median errors before 2000, confirming that inaccuracies are not limited to a few extreme outliers but affect a substantial portion of the dataset.

The visual model follows the metadata on which it was trained and is slightly more robust for most periods except the 1800-1849 period that is just outside the 1850 limit of its training data, and the post-2000 pictures where its performance have been impacted by the overwhelming amount of innacurate data in its training data in that period.

Period	Aggregated	Metadata	Text	Visual
<1800	0.0	268.5	0.0	199.0
1800–1849	0.0	22.0	0.0	75.0
1850–1899	0.0	73.0	0.0	29.0
1900–1949	0.0	79.5	0.0	11.0
1950–1999	0.0	23.0	0.0	9.5
≥2000	0.0	0.0	0.0	5.0

Table 10: Median Absolute Error

5.4.4. 90th Percentile Absolute Error

The 90th percentile absolute error shown in Table 11 evaluates the robustness of each method by focusing on the largest remaining errors.

The aggregated result achieves a value of 0 years for all periods. This indicates that at least 90% of the evaluated pictures were dated exactly correctly. The remaining errors are therefore concentrated within a very small subset of difficult cases.

The metadata field exhibits extremely large values, reaching more than 300 years for pictures before 1800. Such errors are consistent with systematic labeling problems where modern digitization dates replace the actual historical production dates.

The textual extraction method remains highly robust. Except for the period 1950–1999, where a small number of ambiguous descriptions generate mistakes, 90% of the predictions are exact.

The visual model exhibits a similar trend as observed before, where it follows the metadata while being slightly more robust than the metadata, except for the post 2000 era.

Period	Aggregated	Metadata	Text	Visual
<1800	0.0	372.0	0.0	327.6
1800–1849	0.0	200.2	0.0	162.6
1850–1899	0.0	140.0	0.0	61.4
1900–1949	0.0	99.2	0.0	32.2
1950–1999	0.0	57.2	2.5	39.0
≥2000	0.0	0.1	0.0	16.2

Table 11: 90th Percentile Absolute Error

5.4.5. Mean Error (Bias)

The bias values reported in Table 12 reveal systematic tendencies of the different methods.

The aggregated result remains nearly unbiased across all historical periods. The bias never exceeds approximately five years in absolute value and is usually close to zero. This indicates that the aggregation algorithm does not systematically favour older or newer dates.

The metadata field consistently overestimates the date of older pictures. This is particularly visible before 1900 where large positive biases indicate that many images are assigned dates that are far too recent. This observation is consistent with the scanned-date problem identified during the exploratory analysis.

The textual extraction method tends to underestimate dates during the nineteenth and twentieth centuries. This likely occurs because descriptions often mention earlier historical events, publication dates, or birth dates that are mistaken for the creation date of the image.

The visual model exhibits smaller but similar trends. It tends to predict slightly newer dates for older material and slightly older dates for recent material, reflecting a regression-to-the-mean effect common in regression models.

Period	Aggregated	Metadata	Text	Visual
<1800	0.19	259.23	1.53	230.96
1800–1849	0.00	87.45	0.01	82.95
1850–1899	0.97	54.34	−12.88	30.65
1900–1949	1.01	4.67	−20.63	3.43
1950–1999	−0.40	20.32	−25.07	−8.27
≥2000	−5.11	−4.58	−6.23	−12.00

Table 12: Mean Error (Bias)

5.4.6. Median Error (Bias)

The median bias values shown in Table 13 confirm the conclusions drawn from the mean bias analysis.

The aggregated result has a median bias of 0 years for every period, demonstrating the absence of systematic directional errors for the majority of the data.

The textual extraction method also exhibits a median bias of 0 years throughout the benchmark, indicating that its occasional mistakes are not strongly skewed toward either earlier or later dates.

The metadata field and visual model display larger directional tendencies in some periods, further illustrating the value of combining multiple independent information sources.

Period	Aggregated	Metadata	Text	Visual
<1800	0.0	268.5	0.0	199.0
1800–1849	0.0	22.0	0.0	75.0
1850–1899	0.0	0.0	0.0	29.0
1900–1949	0.0	0.0	0.0	4.0
1950–1999	0.0	18.0	0.0	−1.5
≥2000	0.0	0.0	0.0	−4.0

Table 13: Median Error (Bias)

5.4.7. Coverage

Coverage results are reported in Table 14.

The aggregated result achieves 100% coverage across all periods while maintaining the best overall accuracy. This demonstrates that combining multiple sources of information not only improves quality but also avoids sacrificing data availability.

The metadata field naturally reaches 100% coverage because every evaluated picture contains a `date_taken` value. However, the previous results show that this coverage comes at the cost of substantial inaccuracies.

The textual extraction method reaches between 84% and 100% coverage depending on the period. Missing predictions usually correspond to images whose textual fields contain no usable date information.

The visual model provides almost complete coverage because a prediction can be generated from the image itself. This makes it particularly valuable for images lacking textual metadata. The lacking images are sometimes cases where the download of the image timed-out during batch processing.

Period	Aggregated	Metadata	Text	Visual
<1800	100	100	100	99
1800–1849	100	100	100	95
1850–1899	100	100	100	97
1900–1949	100	100	91	100
1950–1999	100	100	96	100
≥2000	100	100	84	100

Table 14: Coverage (%)

5.4.8. Data availability

The Flico dataset is distributed as a PostgreSQL database¹³ composed of four tables: `institution`, `license`, `photo`, and `machine_learning_photo`. The `institution`, `license`, and `photo` tables contain the metadata retrieved from the Flickr API without modification. The `machine_learning_photo` table contains one row per picture and stores all information generated during the processing pipeline described in this study.

In addition to the dating-related fields discussed in this paper, the database also contains fields generated by parallel research efforts focusing on image geolocation and building matching. These fields are distributed together with the dataset to facilitate future work.

Table 15 summarizes the most relevant computed fields available in the `machine_learning_photo` table.

The distribution of intermediate predictions, probabilities, and embeddings makes it possible to reproduce the dating pipeline, evaluate alternative aggregation strategies, or train new models without recomputing expensive image representations.

¹³Zenodo link: <https://zenodo.org/records/20834988>

Field	Description
The following fields are part of this work on date correction	
sig_lip_vect_n	768-dimensional SigLip2 embedding computed from the 320px image used throughout the visual dating pipeline.
is_date_train	Boolean flag indicating whether the picture was part of the training set used to fit the visual regression model.
is_date_test	Boolean flag indicating whether the picture was part of the benchmark used to evaluate the visual regression model.
reg_n_pred_date	Year predicted by the final SVR($C = 50$) visual regression model.
qwen3_pred_date	Year predicted by Qwen3-VL-235B-A22B-Thinking during the visual benchmark.
descr_pred_date	Most probable year extracted from textual metadata (title, description, and tags).
descr_pred_date_1	Second most probable year extracted from textual metadata.
descr_pred_date_2	Third most probable year extracted from textual metadata.
p_descr_date	Probability associated with the first textual candidate.
p_descr_date_1	Probability associated with the second textual candidate.
p_descr_date_2	Probability associated with the third textual candidate.
human_pred_date	Manually annotated year when available as part of the evaluation benchmark.
corrected_year	Final year assigned by the aggregation algorithm and used as the primary date label in Flico.
The following fields are generated by parallel research on geolocation	
clip_vect_224	512-dimensional OpenCLIP embedding computed from a square crop of the image.
geo_cluster_id	Identifier of the geographical cluster associated with the image. Clusters group visually related buildings located in the same area.
geo_group_id	Identifier of a group of visually matching pictures representing the same object or location.
is_central	Flag indicating the representative image selected for a visual group.
mapillary_id	Identifier of the best matching image found in Mapillary.
p_match	Confidence score associated with the Flickr–Mapillary visual match.
is_building	Prediction indicating whether the image depicts a building.
p_building	Probability that the image depicts a building.

Table 15: Computed fields distributed in the machine_learning_photo table.

6. Discussion

The results demonstrate that the main challenge in historical image dating is not necessarily the prediction model itself, but the quality of the available labels. The initial assumption that the Flickr `date_taken` field could serve as ground truth proved incorrect. Large systematic errors were identified, especially for scanned historical material where the digitization date was recorded instead of the creation date.

The textual extraction approach emerged as the strongest individual source of information. Across most historical periods, it achieved near-perfect median accuracy. This highlights an important observation: while institutions do not always populate Flickr metadata consistently, they often preserve accurate historical information inside titles, descriptions, and tags.

The visual model provides complementary information rather than competing directly with textual extraction. Its performance is substantially weaker than text-based extraction. It outperforms a state-of-the-art multimodal reasoning model on a selected benchmark, that proved to be inaccurate. This highlights that the method has potential, but the conclusion that this specific classifier outperforms Qwen in general is wrong.

The aggregation algorithm successfully combines these complementary sources. The resulting labels achieve near-perfect performance on the manually annotated benchmark while maintaining good coverage. This suggests that simple interpretable rules can be highly effective when multiple partially reliable sources of evidence are available.

The main point of this study is that data is often partially incorrect and combining multiple inaccurate sources can be a very efficient approach. The author encourages future work aimed at improving data quality by comparing the label from the Flico dataset with predictions by other models like Qwen or the classifier proposed by Molina, et al. [2].

The author didn't implement such methods to label Flico pictures to stay limited to information from Flickr Commons. Using other sources than the data in Flickr commons to label the picture introduces biases that can make future comparisons more challenging to interpret.

6.0.1. Limitations

Several limitations should nevertheless be acknowledged. First, the final benchmark contains only 600 manually annotated images, offering little guarantee about the actual label quality. Although stratified across time periods, this remains a relatively small sample compared to the nearly two million images contained in the dataset. Second, the aggregation rules were designed through iterative manual observations and may not generalize to future data sources. Third, the visual classifier being trained on the `date_taken` metadata limits its abilities to correct the metadata.

6.0.2. Addressing Limitations

6.0.2.1. Limited Label Quality Guarantee

It is possible to restrict the dataset to 864,395 pictures, that are very likely to be correctly labeled. This corresponds to applying a stricter aggregation algorithm that only considers pictures where the `date_taken` metadata and the best textually extracted candidate year match exactly. This dataset has a distribution that spans 1925-2026 and can be seen in Figure 12. It is the intersection of the distribution of the best textual candidate (Figure 13) and

the distribution of `date_taken` (Figure 14). This requirement is a stronger guarantee of quality as in DEW: remember that DEW is built on the only requirement that the picture's year (metadata field) appears in any text field (see Section 2.2).

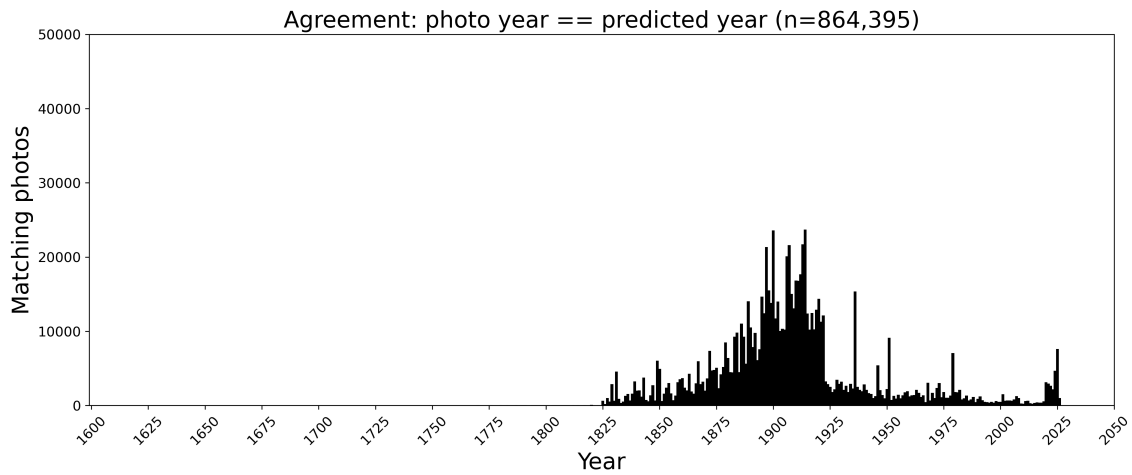


Figure 12: Date distribution of the perfect metadata-textual agreement

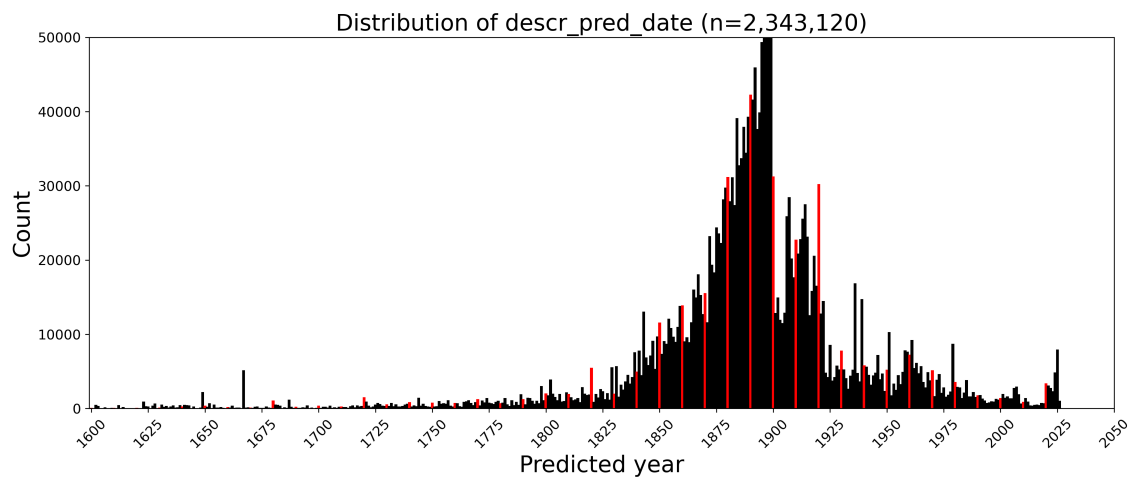


Figure 13: Date distribution per year of the textual candidate

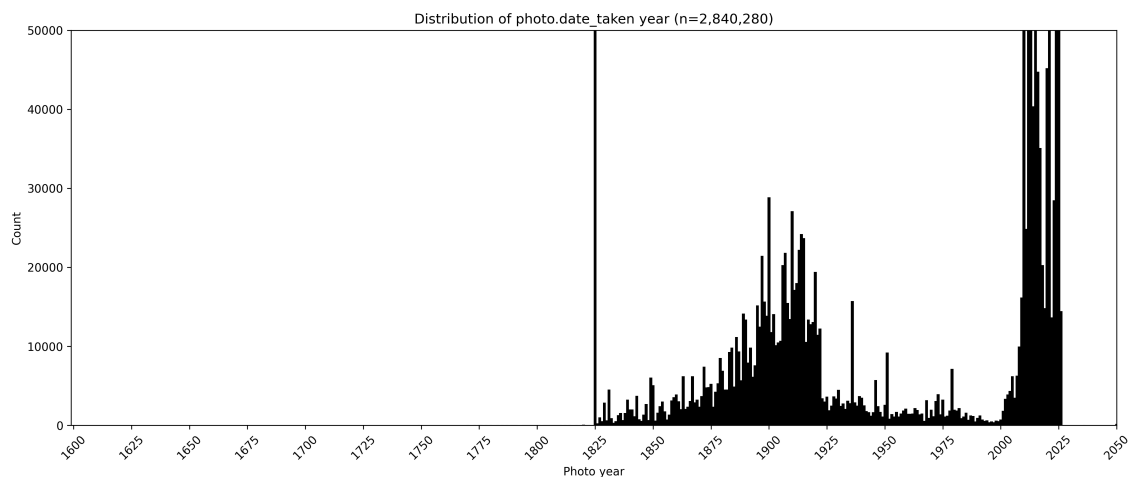


Figure 14: Date distribution per year of the metadata `date_taken` field

6.0.2.2. Methods Won't Generalize to Other Datasets

A lot of rules about the textual extraction are specific to the data observed in this dataset. They are very likely to generalize to future pictures added to Flickr Commons coming from the same institutions, providing that the same annotation conventions are maintained (putting publication dates in brackets, putting the date on a single line, having field names in the description). Pictures from other sources might require new rules, but they are likely to introduce other sources of systematic errors. Identifying these new errors and accounting for them through rules might be more time consuming, less scalable than automatic methods but is interpretable and can be criticized. The author recognizes that more scalable and automatic methods are welcomed, but machine-learning approaches rely on annotated data. This study aims at improving the quality of historical annotated data that is often a blind spot in modern visual models.

6.0.2.3. Visual Classifier: Garbage In/Garbage Out

The trained visual classifier for this study exists for historical reasons: in the beginning of the research, the author believed that the metadata field filtered with the flag `is_date_unknown = false` was accurate enough. The results of the training were helpful to identify the scale of the scanned date error, and that it was not only a problem for the British Library. Other cases highlight further errors in the `date_taken`, that go beyond the scanned-date problem and concern pictures from before 2000. Of course, it is of little value on its own as a predictor of date for random pictures. This classifier plays a relatively minor role in the aggregation algorithm for Flico. For example the picture shown in Figure 15 was taken in 1890. This information is present in the title and is perfectly extracted by the textual algorithm. However, the `date_taken` indicated is 1930. The visual predictor predicts 1926 (probably under the influence of the inaccurate metadata). But this doesn't affect the final result of the label in Flico that is 1890. When a single date is present in the title, it is very likely to be the correct date. The metadata and visual predictor come in handy when multiple dates or no dates are extracted from the textual fields.

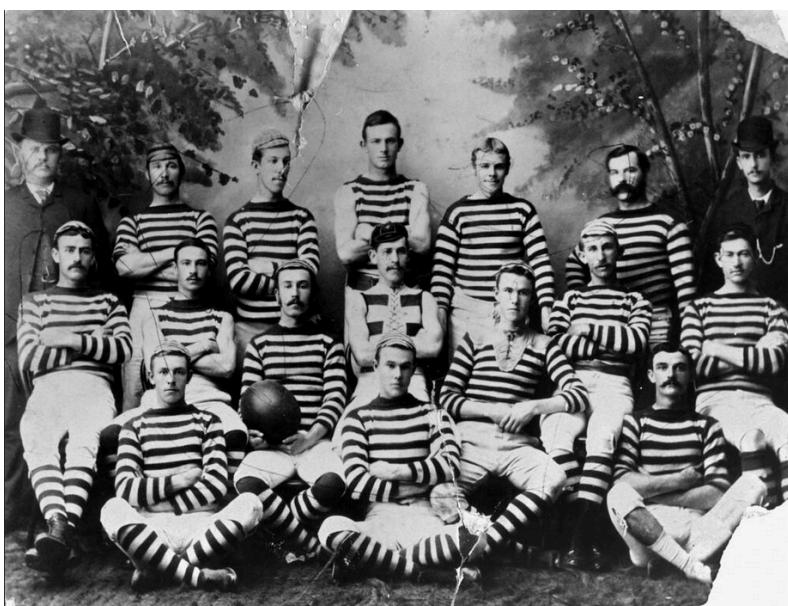


Figure 15: Rockhampton Rugby Union Club, 1890¹⁴

¹⁴<https://www.flickr.com/photos/32605636@N06/27247106097>

7. Conclusion

This work introduced Flico, a dataset of 1,893,415 openly accessible historical images from Flickr Commons with automatically corrected year annotations. Unlike previous datasets, Flico combines metadata, textual extraction, and visual prediction to identify and correct labeling errors rather than simply filtering uncertain examples.

The evaluation demonstrates that the original metadata contains substantial systematic errors, particularly for digitized historical material. A rule-based textual extraction method was shown to recover highly accurate date information from titles, descriptions, and tags. A visual dating model based on SigLip embeddings and Support Vector Regression provides an additional source of evidence. Combining these sources through a simple aggregation strategy produces highly accurate labels while preserving full dataset coverage.

Beyond the corrected year labels, Flico also distributes the intermediate predictions and SigLip embeddings used during the dating process. This significantly increases the long-term value of the dataset. By filtering pictures where the `date_taken` and the best textual candidate differ, the dataset can be filtered down to 864,395 pictures for stronger label quality guarantee. The embeddings can be reused to build visual retrieval systems, similarity search engines, clustering algorithms, or other computer vision applications without recomputing image features. In particular, the combination of corrected dates and visual embeddings could be used to build a historical image search engine allowing users to retrieve visually similar pictures and filter results by time period.

Future work could focus on expanding the manually annotated benchmark, replacing hand-crafted aggregation rules with learned probabilistic models, and exploiting multimodal foundation models specifically trained on historical material. More broadly, the methodology developed here suggests that historical datasets should not merely discard uncertain data, but instead attempt to recover and validate information through the combination of multiple complementary sources.

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